THE FOURIER ENTROPY-INFLUENCE INEQUALITY FOR SYMMETRIC BOOLEAN FUNCTIONS WITH CONSTANT 6.2

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ABSTRACT. We prove that every symmetric Boolean function f satisfies $H[\widehat{f}^2] \le C\, I[f]$ with $C = \log_2(4+e) + 2 + 1/\ln 2 = 6.1908\ldots$, where H is the Fourier entropy in bits and I is the total influence. The previous constant for this class is 12.04, due to O'Donnell, Wright and Zhou. Since f is symmetric, the entropy splits into the entropy of the level weights and a within-level term $B = \sum_k W_k \log_2 \binom{n}{k}$. The level part is at most $\log_2(4+e)\, I$ for every Boolean function. The new part is the bound $B < (2 + 1/\ln 2)\, I$. It rests on a noise-stability estimate for the derivative of a symmetric function, $\mathrm{Stab}_\lambda[D_n f] \le e^{-v} I_0(v)\, \|D_n f\|_2^2$ with $v = (n-1)(1-\lambda^2)/4$, which is a factor of two sharper in the relevant regime than the estimate used by O'Donnell, Wright and Zhou, together with an exact evaluation of the resulting harmonic integral. On the other side, the AND function has $H/I = 4 - (4 - 2/\ln 2)/n + O(2^{-n})$, so the best constant for symmetric functions lies between 4 and 6.1908.

1. INTRODUCTION

Let $f\colon \{-1,1\}^n \to \{-1,1\}$, with Fourier expansion $f = \sum_{S\subseteq[n]} \widehat{f}(S)\chi_S$ under the uniform measure. The Fourier entropy and total influence are

$$H[\widehat{f}^2] = \sum_S \widehat{f}(S)^2 \log_2 \frac{1}{\widehat{f}(S)^2}, \qquad I[f] = \sum_S |S|\, \widehat{f}(S)^2.$$

The Fourier entropy-influence conjecture of Friedgut and Kalai [2] asserts that $H[\widehat{f}^2] \le C\, I[f]$ for a universal constant C. It is open in general. O'Donnell, Wright and Zhou [4] proved it for symmetric functions with $C = 12.04$, and that has remained the constant for the whole symmetric class; later discussions of the symmetric case, for instance [1, 3], still quote it. Here we lower it to 6.1908.

Theorem 1.1. *For every $n \ge 1$ and every symmetric $f\colon \{-1,1\}^n \to \{-1,1\}$,*

$$H[\widehat{f}^2] \le \Big(\log_2(4+e) + 2 + \frac{1}{\ln 2}\Big) I[f] = 6.19078735\ldots\, I[f].$$

Write $W_k = \sum_{|S|=k} \widehat{f}(S)^2$, so that $\sum_k W_k = 1$ and $I = \sum_k k W_k$. When f is symmetric all coefficients at level k have the same square $W_k/\binom{n}{k}$, and therefore

$$H[\widehat{f}^2] = L + B, \qquad L = \sum_k W_k \log_2 \frac{1}{W_k}, \qquad B = \sum_k W_k \log_2 \binom{n}{k}. \tag{1}$$

Theorem 1.1 is the sum of two bounds. Proposition 2.1 gives $L \le \log_2(4+e)\, I$ for every Boolean function; it is Lemma 2.1 of [5], and we include its short proof. Theorem 5.2 gives $B < (2+1/\ln 2)\, I$ for every symmetric Boolean function, and this is the new ingredient.

O'Donnell, Wright and Zhou also bound B through the derivative $D_n f$ and its noise stability; their Theorem 6 gives $\mathrm{Stab}_{1-t}[D_n f] \le (2/\sqrt{\pi})(Nt)^{-1/2} \|D_n f\|_2^2$ with $N = n-1$. We replace it by Proposition 3.2, $\mathrm{Stab}_\lambda[D_n f] \le e^{-v} I_0(v) \|D_n f\|_2^2$ with $v = N(1-\lambda^2)/4$. For $\lambda = 1-t$ with t small and Nt large, $e^{-v} I_0(v) \sim (2\pi v)^{-1/2} \sim (\pi N t)^{-1/2}$, half of the earlier bound. The second step is to keep the stability estimate inside the exact harmonic integral for B rather than integrating a cruder majorant. Two Bessel inequalities (Lemma 4.1) then give the integral in closed form up to terms of order $N^{-1/2}$ with the right sign.

Section 6 records what is known on the other side. The AND function gives $H/I \to 4$ from below, and in an exhaustive computation over all symmetric functions with $n \le 14$ the largest ratio is attained by AND or OR. The within-level part alone approaches $2/\ln 2$ for majority, against our upper bound $2 + 1/\ln 2$. Whether $H \le 4I$ for all symmetric functions is open.

Notation. $N = n - 1$. The derivative in the last coordinate is $D_n f(x) = \frac{1}{2}\big(f(x,1) - f(x,-1)\big)$ for $x \in \{-1,1\}^N$, so $\widehat{D_n f}(T) = \widehat{f}(T \cup \{n\})$ and $\|D_n f\|_2^2 = I_n[f] = I[f]/n$ for symmetric f. For $\lambda \in [0,1]$, $\mathrm{Stab}_\lambda[g] = \sum_T \lambda^{|T|}\widehat{g}(T)^2 = \mathbb{E}[g(x)g(y)]$, where y is obtained from x by keeping each coordinate with probability $(1+\lambda)/2$ and flipping it otherwise. We write

$$F(x) = e^{-x} I_0(x) = \frac{1}{\pi}\int_0^\pi e^{-x(1-\cos\theta)}\,d\theta, \qquad G(x) = e^{-x} I_1(x) = \frac{1}{\pi}\int_0^\pi e^{-x(1-\cos\theta)}\cos\theta\,d\theta,$$

$H_m = \sum_{i=1}^m 1/i$ ($H_0 = 0$), γ for Euler's constant and $E_1(x) = \int_x^\infty e^{-u}u^{-1}\,du$. Logarithms are natural unless written $\log_2$.

2. The level entropy

Proposition 2.1 ([5, Lemma 2.1])**.** *For every Boolean function f, $L = \sum_k W_k \log_2(1/W_k) \le \log_2(4+e)\, I[f]$.*

Proof. If f is constant both sides vanish, so let $\varepsilon = 1 - W_0 > 0$ and $Q_k = W_k/\varepsilon$ for $k \ge 1$. Grouping gives $L = h_2(\varepsilon) + \varepsilon H(Q)$, where h_2 is the binary entropy in bits. Put $a = \log_2(1 + e/4)$ and $P_k = (2^a - 1)2^{-ak}$ for $k \ge 1$, a probability law on the positive integers. Gibbs' inequality gives $H(Q) \le \sum_k Q_k \log_2(1/P_k) = a\,\mathbb{E}_Q k - \log_2(2^a - 1)$, and $\varepsilon \mathbb{E}_Q k = I$. Also $h_2(\varepsilon) \le \varepsilon\log_2(1/\varepsilon) + \varepsilon\log_2 e$. Since $2^a - 1 = e/4$,

$$L \le aI + \varepsilon\big[\log_2(1/\varepsilon) + \log_2 e - \log_2(e/4)\big] = aI + \varepsilon\big[\log_2(1/\varepsilon) + 2\big].$$

It remains to show $\varepsilon[\log_2(1/\varepsilon) + 2] \le 2I$. Choose the sign of f so that $p = \Pr[f = -1] \le 1/2$. Then $\varepsilon = 1 - (1-2p)^2 = 4p(1-p)$, and the edge-isoperimetric inequality of the cube gives $I \ge 2p\log_2(1/p)$. A direct expansion shows

$$2p\log_2\frac{1}{p} - \frac{\varepsilon}{2}\Big[\log_2\frac{1}{\varepsilon} + 2\Big] = 2p\big[p\log_2(1/p) + (1-p)\log_2(1-p)\big].$$

The bracket $\varphi(p)$ vanishes at $p = 0$ and $p = 1/2$, and $\varphi''(p) = (2p-1)/(p(1-p)\ln 2) \le 0$ on $(0, 1/2]$, so $\varphi \ge 0$ there. Hence $L \le (a+2)I = \log_2(4+e)\, I$. □

3. Noise stability of the derivative

Lemma 3.1. *Let (a_w) be a finite nonnegative unimodal sequence and let $\epsilon_w \in \{-1, 0, 1\}$ be such that the nonzero ϵ_w alternate in sign as w increases. Then $\big|\sum_w \epsilon_w a_w\big| \le \max_w a_w$.*

Proof. Write $\sum_w \epsilon_w a_w = \int_0^\infty \sum_{w:\, a_w > u} \epsilon_w\,du$. Each set $\{w : a_w > u\}$ is an interval, so the inner sum is an alternating sum of signs over consecutive nonzero entries, which is 0 or ± 1, and it vanishes for $u \ge \max_w a_w$. □

Proposition 3.2. *Let f be a symmetric Boolean function on $n = N + 1$ variables and $g = D_n f$. For every $\lambda \in [0,1]$,*

$$\mathrm{Stab}_\lambda[g] \le F(v)\,\|g\|_2^2, \qquad v = \frac{N(1-\lambda^2)}{4}.$$

Proof. The function g is symmetric with values in $\{-1, 0, 1\}$, so $g(x) = g_{|x|}$ where $|x|$ counts the coordinates equal to -1. As the weight increases, f changes value exactly at the layers where $g \ne 0$, and consecutive changes of a ± 1-valued function have opposite signs; so the nonzero g_w alternate. Condition on $|x| = k$. The weight $|y|$ is then a sum of N independent Bernoulli variables, k of them

with mean $(1+\lambda)/2$ and $N-k$ with mean $(1-\lambda)/2$, each with variance $(1-\lambda^2)/4$. Its law q_k is a Poisson-binomial law, hence log-concave and unimodal, and by Lemma 3.1

$$\big|\mathbb{E}[g(y) \mid |x|=k]\big| = \Big|\sum_w q_k(w) g_w\Big| \le \max_w q_k(w).$$

A Bernoulli(p) variable has characteristic function of modulus

$$\big(1-4p(1-p)\sin^2(\theta/2)\big)^{1/2} \le \exp\big(-2p(1-p)\sin^2(\theta/2)\big),$$

so the characteristic function of q_k has modulus at most $\exp(-2v\sin^2(\theta/2)) = \exp(-v(1-\cos\theta))$. By Fourier inversion every atom of q_k is at most $\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-v(1-\cos\theta)}d\theta = F(v)$. Therefore

$$\mathrm{Stab}_\lambda[g] = \sum_k \Pr[|x|=k]\, g_k\, \mathbb{E}[g(y) \mid |x|=k] \le F(v) \sum_k \Pr[|x|=k]\, |g_k| = F(v)\, \mathbb{E}|g| = F(v)\, \|g\|_2^2,$$

using $|g| = g^2$. □

4. Two Bessel inequalities

Lemma 4.1. (a) $\sqrt{w}\,F(w) > 1/\sqrt{2\pi}$ *for every* $w \ge 1/2$.
(b) $J(X) := \int_0^X u\big(F(u)-G(u)\big)\,du = X\,G(X)$ *for* $X \ge 0$.
(c) $G(X) < 1/\sqrt{2\pi X}$ *for every* $X > 0$.

Proof. (a) Let $h(w) = \sqrt{w}\,F(w)$. Substituting $I_0 = e^w F$ into $I_0'' + I_0'/w - I_0 = 0$ gives $h'' + 2h' + h/(4w^2) = 0$, so at any stationary point $h'' = -h/(4w^2) < 0$, and h has no local minimum in $(1/2,\infty)$. At the left end, $I_0(1/2) \ge 1$ gives $h(1/2) \ge e^{-1/2}/\sqrt{2} > 1/\sqrt{2\pi}$, since $\pi > e$. At infinity, substituting $\theta = y/\sqrt{w}$,

$$h(w) = \frac{1}{\pi}\int_0^{\pi\sqrt{w}} \exp\big[-w\big(1-\cos(y/\sqrt{w})\big)\big]\,dy \longrightarrow \frac{1}{\pi}\int_0^\infty e^{-y^2/2}\,dy = \frac{1}{\sqrt{2\pi}},$$

by dominated convergence with the majorant e^{-2y^2/π^2} (from $1-\cos\theta \ge 2\theta^2/\pi^2$ on $[0,\pi]$). If $h(w_0) < 1/\sqrt{2\pi}$ for some $w_0 > 1/2$, choose $R > w_0$ with $h(R) > h(w_0)$; the minimum of h on $[1/2, R]$ is then attained at an interior point, which would be a local minimum. So $h \ge 1/\sqrt{2\pi}$ on $[1/2,\infty)$, and a point of equality would again be an interior local minimum.

(b) From the integral representations, $F' = G - F$ and $G' = F - G - G/x$. Hence $(xG)' = x(F-G)$, and $xG(x) \to 0$ as $x \to 0$.

(c) Integrating by parts in θ, $G(X) = \frac{X}{\pi}\int_0^\pi e^{-X(1-\cos\theta)}\sin^2\theta\,d\theta$, and the substitution $u = 1-\cos\theta$ gives

$$G(X) = \frac{X}{\pi}\int_0^2 e^{-Xu}\sqrt{u(2-u)}\,du < \frac{\sqrt{2}\,X}{\pi}\int_0^\infty e^{-Xu}\sqrt{u}\,du = \frac{1}{\sqrt{2\pi X}}.$$ □

5. The within-level entropy

For $N \ge 1$ define

$$C_N = \int_0^1 \frac{F\big(Nt(2-t)/4\big) - (1-t)^N}{t}\,dt.$$

The integrand is bounded near $t = 0$, because both terms equal $1 - O(t)$.

Proposition 5.1. $C_N < \ln 4$ *for every integer* $N \ge 1$.

Proof. Split $C_N = T_1 + T_2 + T_3$ with

$$T_1 = \int_0^1 \frac{F(Nt(2-t)/4) - F(Nt/2)}{t}\,dt, \qquad T_2 = \int_0^1 \frac{F(Nt/2) - e^{-Nt}}{t}\,dt,$$

and $T_3 = \int_0^1 \big(e^{-Nt} - (1-t)^N\big)t^{-1}\,dt$.

The term T_2. Write $F(x) = \mathbb{E}_\theta e^{-x(1-\cos\theta)}$ with θ uniform on $[0,\pi]$. By Tonelli and Frullani's integral,

$$\int_0^\infty \frac{F(Nt/2) - e^{-Nt}}{t}\,dt = \mathbb{E}_\theta \ln\frac{2}{1-\cos\theta} = -2\,\mathbb{E}_\theta \ln\sin\frac{\theta}{2} = -\frac{4}{\pi}\int_0^{\pi/2} \ln\sin u\,du = \ln 4.$$

On $t \ge 1$ we have $Nt/2 \ge 1/2$, so Lemma 4.1(a) gives $F(Nt/2) > (\pi Nt)^{-1/2}$ there, and

$$\int_1^\infty \frac{F(Nt/2) - e^{-Nt}}{t}\,dt \ge \frac{1}{\sqrt{\pi N}}\int_1^\infty t^{-3/2}\,dt - E_1(N) = \frac{2}{\sqrt{\pi N}} - E_1(N).$$

Hence $T_2 \le \ln 4 - 2/\sqrt{\pi N} + E_1(N)$.

The term T_1. The function F is completely monotone, so $-F' = F - G$ is positive and decreasing. With $a = Nt(2-t)/4$ and $b = Nt/2$ we have $Nt/4 \le a \le b$ and $b - a = Nt^2/4$, so $F(a) - F(b) \le (b-a)\big(-F'(Nt/4)\big)$. Substituting $u = Nt/4$ and using Lemma 4.1(b),

$$T_1 \le \int_0^1 \frac{Nt}{4}\big(-F'(Nt/4)\big)\,dt = \frac{4}{N}\,J(N/4) = G(N/4) < \sqrt{\frac{2}{\pi N}}$$

by Lemma 4.1(c).

The term T_3. Since $\int_0^1 \big(1-(1-t)^N\big)t^{-1}dt = H_N$ and $\int_0^1 \big(1-e^{-Nt}\big)t^{-1}dt = \int_0^N (1-e^{-u})u^{-1}du = \ln N + \gamma + E_1(N)$,

$$T_3 = H_N - \ln N - \gamma - E_1(N) \;\ge 0.$$

Adding the three estimates, the E_1 terms cancel and

$$C_N < \ln 4 - \frac{2-\sqrt{2}}{\sqrt{\pi N}} + \big(H_N - \ln N - \gamma\big) < \ln 4 - \frac{2-\sqrt{2}}{\sqrt{\pi N}} + \frac{1}{2N}.$$

The last two terms sum to a negative number when $N > \pi/\big(4(2-\sqrt{2})^2\big)$, and this threshold is below 3 because $\pi < 4 < 12(2-\sqrt{2})^2$ (the last inequality follows from $\sqrt{2} < 17/12$). So $C_N < \ln 4$ for $N \ge 3$.

For $N = 1$, $F \le 1$ gives $C_1 \le \int_0^1 \big(1-(1-t)\big)t^{-1}dt = 1 < \ln 4$. For $N = 2$ put $\lambda = 1-t$ and $v = (1-\lambda^2)/2$. From $e^{-z} \le 1 - z + z^2/2$ for $z \ge 0$ and $\mathbb{E}_\theta(1-\cos\theta) = 1$, $\mathbb{E}_\theta(1-\cos\theta)^2 = 3/2$ we get $F(v) \le 1 - v + \frac{3}{4}v^2$, and

$$C_2 \le \int_0^1 \frac{\frac{1}{2}(1-\lambda^2) + \frac{3}{16}(1-\lambda^2)^2}{1-\lambda}\,d\lambda = \int_0^1 \Big[\frac{1+\lambda}{2} + \frac{3}{16}(1-\lambda)(1+\lambda)^2\Big]d\lambda = \frac{3}{4} + \frac{3}{16}\cdot\frac{11}{12} = \frac{59}{64} < \ln 4.$$

□

Theorem 5.2. *For every symmetric Boolean function f on n variables, $B \le \frac{1+C_{n-1}}{\ln 2}\,I[f] < \Big(2 + \frac{1}{\ln 2}\Big)I[f]$ when $n \ge 2$, and $B = 0$ when $n = 1$.*

Proof. For $n = 1$, $\binom{1}{k} = 1$. Let $n \ge 2$ and assume $I > 0$. Let $g = D_n f$ on $N = n-1$ variables. Since $\widehat{g}(T)^2 = \widehat{f}(T \cup \{n\})^2 = W_{|T|+1}/\binom{n}{|T|+1}$, the level weights of g are $W_j[g] = \binom{N}{j}W_{j+1}/\binom{n}{j+1} = (j+1)W_{j+1}/n$, and $\|g\|_2^2 = I/n$. So $q_j = (j+1)W_{j+1}/I$, $0 \le j \le N$, is a probability law, and

$$\frac{B}{I} = \sum_{j=0}^{N} q_j\,\frac{\log_2\binom{n}{j+1}}{j+1} \le \frac{1}{\ln 2}\Big[1 + \mathbb{E}_q \ln\frac{n}{j+1}\Big],$$

using $\ln\binom{n}{k} \le k\big(1 + \ln(n/k)\big)$. Comparing sums with integrals, $\ln\frac{n}{j+1} \le H_N - H_j = \int_0^1 \frac{\rho^j - \rho^N}{1-\rho}\,d\rho$, with a nonnegative integrand. Let $s(\rho) = \sum_j q_j\rho^j = \mathrm{Stab}_\rho[g]/\|g\|_2^2$. Taking the expectation under q and putting $\rho = 1-t$,

$$\mathbb{E}_q(H_N - H_j) = \int_0^1 \frac{s(1-t) - (1-t)^N}{t}\,dt \le \int_0^1 \frac{F\big(Nt(2-t)/4\big) - (1-t)^N}{t}\,dt = C_N,$$

where Proposition 3.2 with $\lambda = 1-t$ gives $s(1-t) \le F(N(1-(1-t)^2)/4) = F(Nt(2-t)/4)$. Hence $B/I \le (1+C_N)/\ln 2$, and Proposition 5.1 gives $(1+C_N)/\ln 2 < (1+\ln 4)/\ln 2 = 2+1/\ln 2$. □

Proof of Theorem 1.1. Combine (1), Proposition 2.1 and Theorem 5.2: $H[\widehat{f}^2] = L+B \le \big(\log_2(4+e)+2+1/\ln 2\big) I[f]$, and $\log_2(4+e) = 2.748092\ldots$, $2+1/\ln 2 = 3.442695\ldots$. □

Remark 5.3. The values of C_N computed by quadrature are $C_1 = 0.6638$, $C_2 = 0.8911$, $C_3 = 0.9991$, $C_{100} = 1.3114$, $C_{1000} = 1.3616$ and $C_{10^5} = 1.3838$, against $\ln 4 = 1.3863$; $\ln 4 - C_N$ is about $0.8\,N^{-1/2}$ for large N. So the argument cannot give a within-level constant below $2+1/\ln 2$ without a better stability input or a better use of it.

6. The other side

Proposition 6.1. *For the AND function on n variables,*

$$\frac{H[\widehat{f}^2]}{I[f]} = 4 - \frac{4 - 2/\ln 2}{n} + O(2^{-n}).$$

Proof. Let $f = 1 - 2\cdot\mathbf{1}[x = (-1,\ldots,-1)]$. Then $\widehat{f}(\emptyset) = 1-2^{1-n}$ and $\widehat{f}(S)^2 = 4^{1-n}$ for $S \ne \emptyset$, so $I = 4^{1-n}\sum_k k\binom{n}{k} = n2^{1-n}$. The 2^n-1 nonempty sets contribute $(2^n-1)4^{1-n}\cdot 2(n-1) = 8(n-1)2^{-n} + O(4^{-n}n)$ to the entropy, and the empty set contributes $-(1-2^{1-n})^2\log_2(1-2^{1-n})^2 = 2^{2-n}/\ln 2 + O(4^{-n})$. Dividing by I gives $4(n-1)/n + 2/(n\ln 2) + O(2^{-n})$. □

So the AND function approaches the ratio 4 from below, and the best constant for symmetric functions is at least 4. An exhaustive computation over all 2^{n+1} symmetric functions for each $n \le 14$ (Section 7) finds the largest ratio at AND or OR (or a function obtained from them by negating the input or the output) in every case. For the within-level part alone the natural test case is majority: for $n = 2m+1$ the derivative of majority is the indicator of the middle layer, its normalized spectral law is $q_{2j} = \binom{2j}{j}\binom{2m-2j}{m-j}4^{-m}$ (and $q_{2j+1} = 0$), and $B/I \to 2/\ln 2$ as $m \to \infty$. So the best within-level constant is at least $2/\ln 2 = 2.885$, against our bound $2+1/\ln 2 = 3.443$. The best constant for the whole entropy of symmetric functions lies in [4, 6.1908].

7. Numerical checks

None of the proofs above depends on a computation. As independent checks, the repository contains two scripts. The first, at 30 digits, evaluates $h(w) - 1/\sqrt{2\pi}$ on a logarithmic grid of $[1/2, 5\cdot 10^9]$ (the smallest value is $1.1\cdot 10^{-11}$, at the top of the grid, as the limit predicts), checks $J(X) = XG(X)$ and Lemma 4.1(c) at several points and the identity for T_3, and computes C_N by quadrature for $N \le 40$ and $N = 100$, 1000, 10^5 against the bound in the proof of Proposition 5.1. The second computes the spectrum of every nonconstant symmetric Boolean function with $n \le 14$ from the Krawtchouk transform in exact integers, and checks $L \le \log_2(4+e)I$, $B \le (1+C_{n-1})I/\ln 2$ and Theorem 1.1 on each; it also records the function with the largest ratio H/I for each n. The largest ratios found are $L/I = 2.5176$, $B/I = 1.7452$ and $H/I = 3.9201$, the last for AND at $n = 14$.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [6] and working together inside the Brain framework [8], a shared, persistent memory

in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [7], the Math Brain repository [6] and the Brain framework repository [8].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The mathematics, the numerical checks and the manuscript were produced by AI systems. Codex (OpenAI) agents found the decomposition (1) with a first within-level bound, the stability estimate of Proposition 3.2, the proofs of the Bessel inequalities of Lemma 4.1 (correcting a sign error in an earlier outline of Proposition 5.1), the computations of Section 6, and an independent line-by-line review of the full argument with its own numerical checks. A Claude (Anthropic) agent coordinated the work, re-derived each step by hand, ran further checks, and wrote this manuscript and its publication files. The author selected the problem and directed the work; the author did not carry out the mathematics or the writing. Responsibility for submission or publication rests with the author. The AI-assisted internal reviews and compilation checks are not external peer review or formal verification.

Code and data availability

The checks of Section 7 are produced by the scripts in the `code` directory of the repository https://github.com/yspennstate/symmetric-fei-constant. The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [6].

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